week 3

symbol / baud rate $D = \frac{N}{T_0}$ symbols / second （N - 符号数）

bit rate $R = \frac{n}{T_0}$ bits/second （n - 比特数）

binary: $n = N$ $R = D$

bandwidth: $B \geq \frac{N}{2T_0} = \frac{1}{2}D$

↳ 输出数字波形

若为 rectangular pulse shape, $B = \frac{1}{T_s} = D$

若为 $\frac{\sin x}{x}$ type, 可取等 $B = \frac{1}{2}D$

↳ 输出模拟波形

$T_b = \frac{T_0}{n}$ 传递一位所用的时间

二进制转多电平信号:

binary 中: $D = R$ 8 symbols ↔ 8 bits

多电平中: 8 symbols → 4 symbols, 每个 symbol 2 bits

↓ $D = \frac{R}{L} \neq R$

↓ L

$L = 2^l$

↓ l

0 0 0 1 1 0

-3 -1 +1 +3 $N = 4$ $L = 4$

T_s 传递一个 symbol 的时间

T_b 传递一位

T_0 传递总信息(如一串 binary)

bit rate $R = \frac{n}{T_0} = \frac{1}{T_s}$ $\quad B = \frac{1}{2}f_s = \frac{1}{2T_s}$

band rate $D = \frac{N}{T_0} = \frac{1}{T_s}$

digital signal: $s(t) = \sum_{n=-\infty}^{+\infty} a_n f(t - nT_s)$

random data — a_n; symbol pulse shape — $f(t-nT_s)$; duration

- binary $T_s = T_b$
- multilevel $T_s = lT_b$

(line code)

功率谱密度 PSD.

$$P(f) = \frac{|F(f)|^2}{T_s} \sum_{K=-\infty}^{\infty} R(K) e^{j2\pi K f T_s}$$

$F(f)$ 的傅立叶变换

$$R(K) = \sum_{i=1}^{L} (a_n a_{n+K})_i P_i$$

① Unipolar NRZ

$k=0$ 时，$A\times A$，0×0，$I=2$，$A^2:50\%$ $0:50\%$

$$R(0)=\sum_{i=1}^{2}(a_n a_n)_i P_i = A^2\cdot\frac{1}{2}+0^2\cdot\frac{1}{2}=\frac{1}{2}A^2$$

$k\neq 0$ 时，$I=4$，$A\times A$，$A\times 0$，$0\times A$，0×0，各占 $\frac{1}{4}$

$$R(k)=\sum_{i=1}^{4}(a_n a_{n+k})_i P_i = \frac{1}{4}A^2$$

$$R(k)=\begin{cases}\frac{1}{2}A^2 & k=0\\ \frac{1}{4}A^2, & k\neq 0\end{cases}$$

$$P(f)=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2\left[1+\frac{1}{T_b}\sum_{n=-\infty}^{\infty}\delta\left(f-\frac{n}{T_b}\right)\right]$$

考试要写过程，P39

$$=\frac{A^2T_b}{4}\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2\left[1+\frac{1}{T_b}\delta(f)\right]$$

② Polar NRZ

$k=0$ $R(0)=A^2\cdot\frac{1}{2}+(-A)^2\cdot\frac{1}{2}=A^2$

$k\neq 0$ $R(k)=\sum_{i=1}^{4}\cdots = A^2\cdot\frac{1}{4}+(-A)(+A)\cdot\frac{1}{4}+(+A)(-A)\cdot\frac{1}{4}+(-A)^2\cdot\frac{1}{4}=0$

$$R(k)=\begin{cases}A^2, k=0\\ 0, k\neq 0\end{cases}$$

$$P(f)=A^2T_b\left(\frac{\sin\pi fT_b}{\pi fT_b}\right)^2$$

③ unipolar RZ

$$F(f)=\frac{T_b}{2}\cdot\frac{\sin\pi f(T_b/2)}{\pi fT_b/2}$$ R(k) 与 unipolar NRZ 一样

$$P(f)=\frac{A^2T_b}{16}\left(\frac{\sin\pi fT_b/2}{\pi fT_b/2}\right)^2\left[1+\frac{1}{T_b}\sum_{n=-\infty}^{\infty}\delta\left(f-\frac{n}{T_b}\right)\right]$$

T_s

④ bipolar RZ

$k=0$ $R(0)=\frac{1}{2}A^2$

$k=1$ $R(1)=-\frac{A^2}{4}$ $(-A^2,0,0,0)$ $I=4$

$k>1$ $R(k>1)=A^2\frac{1}{8}-A^2\frac{1}{8}=0$ $(\mp A^2,0,0,0)$ $I=8$

$$R(k)=\begin{cases}\frac{A^2}{2}, k=0\\ -\frac{A^2}{4}, |k|=1\\ 0, |k|>1\end{cases}$$

$$F(f) = \frac{T_b}{2} \cdot \frac{\sin \pi f (T_b/2)}{\pi f T_b/2}$$

$$P(f) = \frac{A^2 T_b}{8} \left(\frac{\sin \pi f T_b/2}{\pi f T_b/2}\right)^2 [1 - \cos(2\pi f T_b)]$$

$$= \frac{A^2 T_b}{4} \left(\frac{\sin \pi f T_b/2}{\pi f T_b/2}\right)^2 [\sin^2(\pi f T_b)]$$

⑤ Manchester NRZ

$R(0) = \frac{1}{2}A^2 + \frac{1}{2}A^2 = A^2$, $k=0$

$k \neq 0$ 时, $R(k) = \frac{1}{4}A^2 + \frac{1}{4}(-A^2) + \frac{1}{4}(-A^2) + \frac{1}{4}A^2 = 0$

$$R(k) = \begin{cases} A^2, & k=0 \\ 0, & k \neq 0 \end{cases}$$

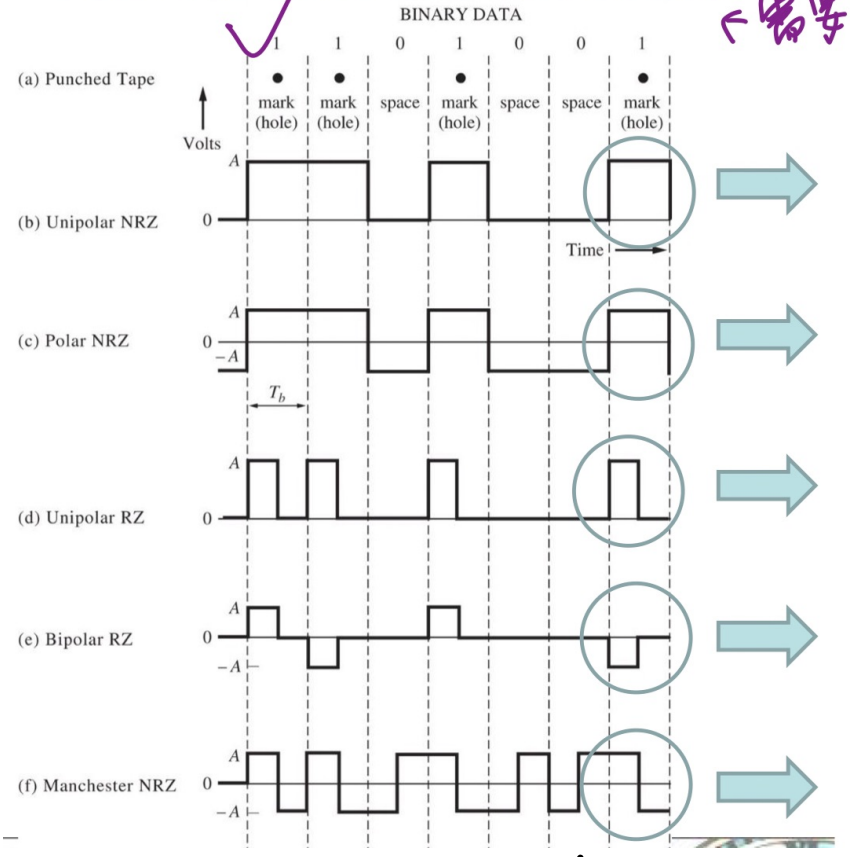
$$P(f) = A^2 T_b \left(\frac{\sin \pi f T_b/2}{\pi f T_b/2}\right)^2 [\sin^2(\pi f T_b/2)]$$

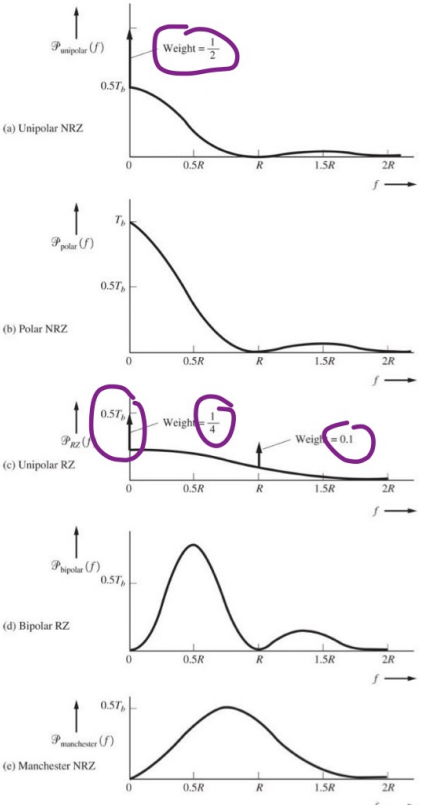
half-width rectangular pulse

Table 3-3 spectral efficiencies of line codes

Code Type		First Null Bandwidth (Hz)	Spectral Efficiency $\eta = R/B$ [(bits/s)/Hz]
Unipolar NRZ	2R	R	1
Polar NRZ	2R	R	1
Unipolar RZ		$2R$	$\frac{1}{2}$
Bipolar RZ	R	R	1
Manchester NRZ		$2R$	$\frac{1}{2}$
Multilevel polar NRZ		R/ℓ	ℓ

PSD for different line codes





1 1 0 1 0 0 1 0 1

spectral efficiency

$\eta = \frac{R}{B}$ → bit rate
→ bandwidth

$\eta_{max} = \frac{C}{B} = \log_2(1+\frac{S}{N})$

对于多电平 polar NRZ: $B_{null} = \frac{R}{L} \Rightarrow \eta = L$

ISI

$H(f) = H_t(f) H_c(f) H_r(f) \longleftrightarrow h(t) = h_t(t) * h_c(t) * h_r(t)$

reduce ISI:

① increase the channel bandwidth

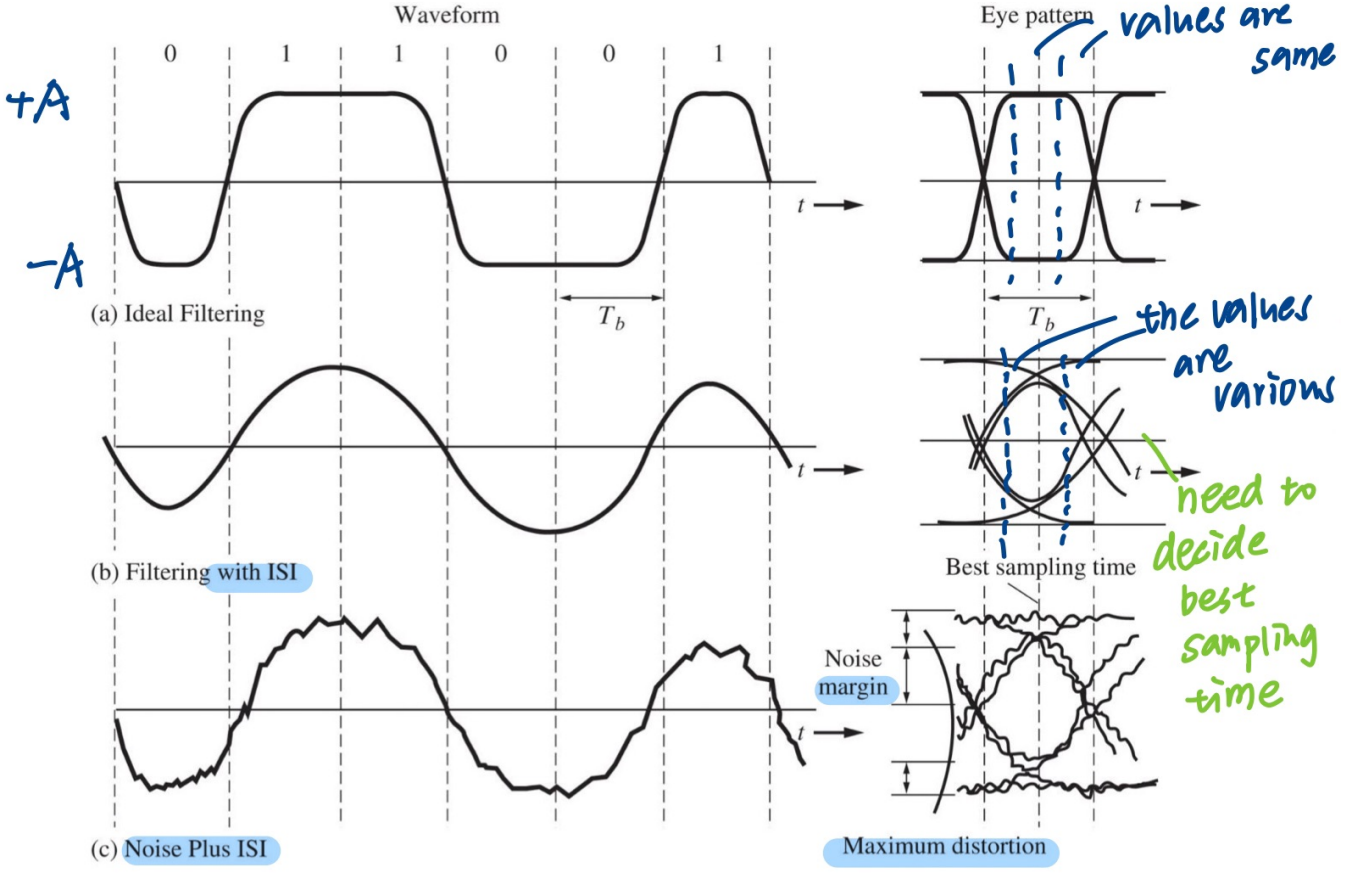
② reshape the pulse from the transmitter

the band rate of the raised cosine-rolloff system with no ISI:

$D = \frac{2B}{1+r}$ → 带宽 absolute bandwidth

↘ system rolloff factor

Distorted polar NRZ waveform and corresponding eye pattern



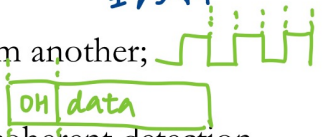
The eye patterns provide the following information

- The *timing error* allowed on the sampler at the receiver is given by the width inside the eye, called the eye opening. Of course, the preferred time for sampling is at the point where the vertical opening of the eye is largest.

- The sensitivity to timing error is given by the slope of the open eye (evaluated at, or near, the zero-crossing point).

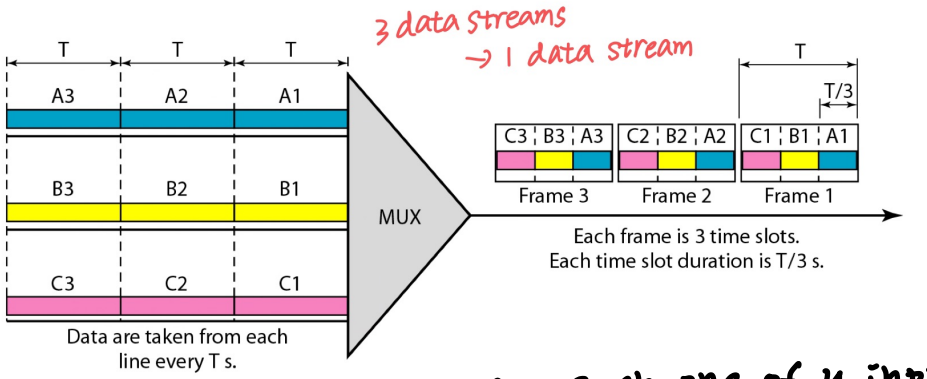
- The noise margin of the system is given by the height of the eye opening

- Digital communications usually need at least three types of synchronisation signals:
  - Bit sync, to distinguish one bit interval form another;
  - Frame sync, to distinguish groups of data;
  - Carrier sync, for bandpass signalling with coherent detection.



For Time-Division Multiplexing

Figure: *Synchronous time-division multiplexing*



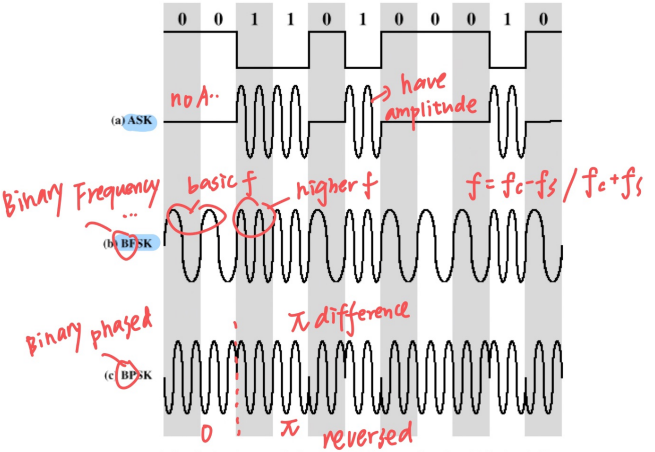
given: x bps data rate for each one of y input connection
1 bit at a time is multiplexed (a unit is 1 bit)

① duration of each input time slot: $\frac{1}{x}$
② duration of each output time slot: $\frac{1}{xy}$
③ duration of each frame: $\frac{1}{x}$
④ output bit rate: xy ⑤ output frame rate: x = input rate

week 4

基本的编码技术

- Digital data to analog signal
  - Amplitude-shift keying (ASK) $S(t)=\begin{cases}A\cos(2\pi f_c t) & 1\\ 0 & 0\end{cases}$
    - Amplitude difference of carrier frequency
  - Frequency-shift keying (FSK) $S(t)=\begin{cases}A\cos(2\pi f_1 t) & 1\\ A\cos(2\pi f_2 t) & 0\end{cases}$
    - Frequency difference near carrier frequency
  - Phase-shift keying (PSK) $S(t)=\begin{cases}A\cos(2\pi f_c t) & 1\\ A\cos(2\pi f_c t+\pi) & 0\end{cases}$
    - Phase of carrier signal shifted



Modulation of Analog Signals for Digital Data

对于MFSK: $S_i(t)=A\cos 2\pi f_i t,\ 1\le i\le M$

$T_s = LT$ (T: bit period), $M=2^L$, $f_i = f_c + (2i-1-M)f_d$ (f_c: carrier f, f_d: difference f)

最小频率间隔 $2f_d = \frac{1}{T_s}$

调制器带宽 $W_d = \frac{2^L}{LT} = \frac{M}{T_s}$

编码 L bits 需要带宽 $2Mf_d$

$R = \frac{L}{T_s} = \frac{\log_2 M}{T_s} = \log_2 M \cdot (2f_d)$

数字调制传输信号: (载波正弦, 数字调制)

$S(t) = Re[(a_k + jb_k)e^{j2\pi f_c t}]$

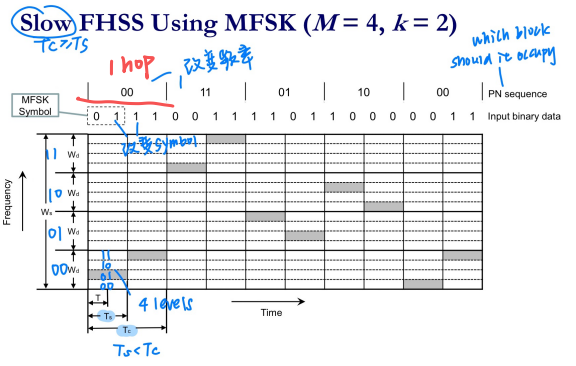
① FHSS 跳频扩频

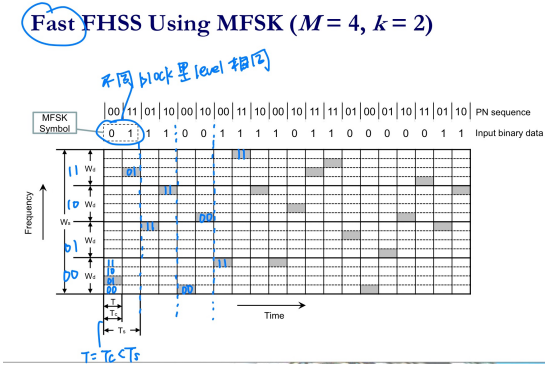
duration of a bit $T = \frac{1}{R}$ ↗ data rate

duration of signal symbol $T_s = LT$ second

T_c：改变频率所需时间

$\begin{cases} T_c \gg T_s & \text{slow} \\ T_c < T_s & \text{fast} \end{cases}$ use MFSK





② DSSS use BPSK

↓

$S_d(t) = A_d(t)\cos(2\pi f_c t)$

Amplitude / DSSS

carrier frequency

modulation

$\Rightarrow S(t) = S_d(t) \cdot C(t)$ sender

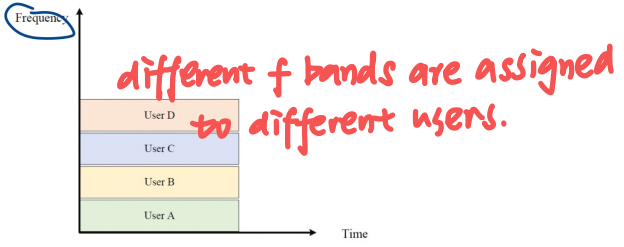
receiver：再次 $\times C(t)$ → recover

多址接入

① FDMA 频分多址

频分多址

- **Key features**:
  - Assign each user to a particular channel.
  - Transmit signals simultaneously and continuously.to enable multiple users to share the same channel simultaneously.



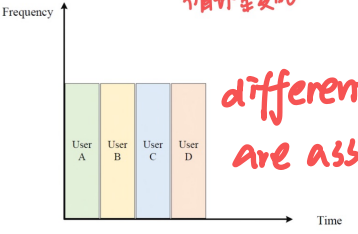
different f bands are assigned to different users.

- **Application**: all 1G systems use FDMA.

② TDMA 时分多址

- **Key features**
  - Single carrier frequency with multiple users.
  - Non-continuous transmission.
  - Each user occupies a **cyclically repeating** time slot.

循环重复的

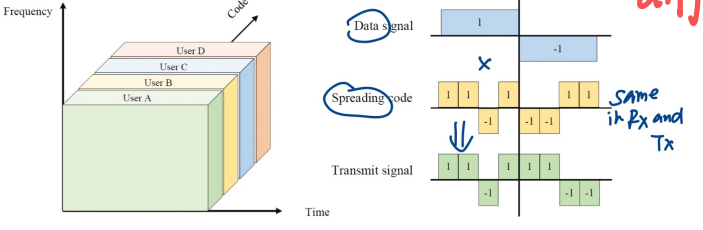


different time slots are assigned to....

- **Application**: most 2G systems use TDMA.

③ CDMA 码分多址

- **Key features**
  - All users use same time and frequency.
  - Narrowband signals multiplied by wideband spreading codes.



different codes are assigned to different users.

then can recover

- **Application**: some 2G and most 3G systems.

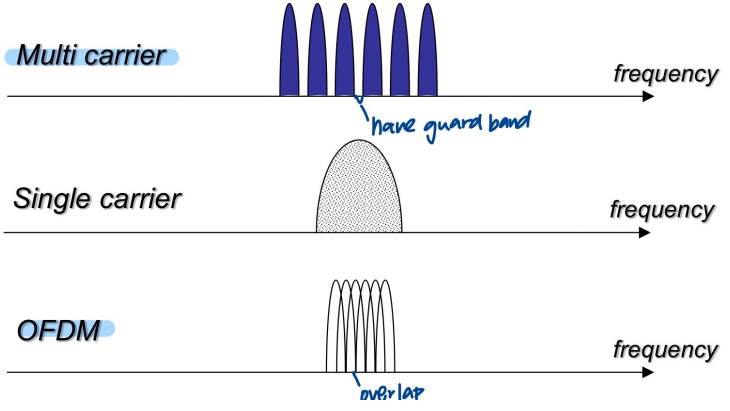
OFDMA

4G and 5G systems

- **OFDMA** = **O**rthogonal **F**requency **D**ivision **M**ultiple **A**ccess
- OFDMA is a multi-user version of the popular OFDM digital modulation scheme. Multiple access is achieved in OFDMA by assigning subsets of subcarriers to individual users.

Same data rate transmission

Spectrum comparison for same data rate transmission



OFDM vs. Multi carrier

- OFDM is a multi-carrier modulation
- OFDM sub-carrier spectrum is overlapping
- In FDMA, band-pass filter separates each transmission
- In OFDM, each sub-carrier is separated by DFT because carriers are orthogonal
  - Condition of the orthogonality will be explained later
- Each sub-carrier is modulated by PSK, QAM

Thousands of PSK/QAM symbol can be simultaneously transmitted in one OFDM symbol

基带OFDM信号

$$s_B(t) = \sum_{n=0}^{N-1} \{a_n \cos(2\pi n f_0 t) - b_n \sin(2\pi n f_0 t)\}$$

$$\frac{T}{2} a_k = \int_0^T s_B(t) \cdot \cos(2\pi k f_0 t)\, dt$$

$$\frac{T}{2} b_k = \int_0^T s_B(t) \{-\sin(2\pi k f_0 t)\} dt$$

OFDM产生：

OFDM signal generation in digital domain

step1: ◆ Define complex base-band signal u(t) as follows

$$s_B(t) = \mathrm{Re}[u(t)]$$

$$u(t) = \sum_{n=0}^{N-1} d_n \cdot e^{j2\pi n f_0 t}, \quad d_n = a_n + jb_n$$

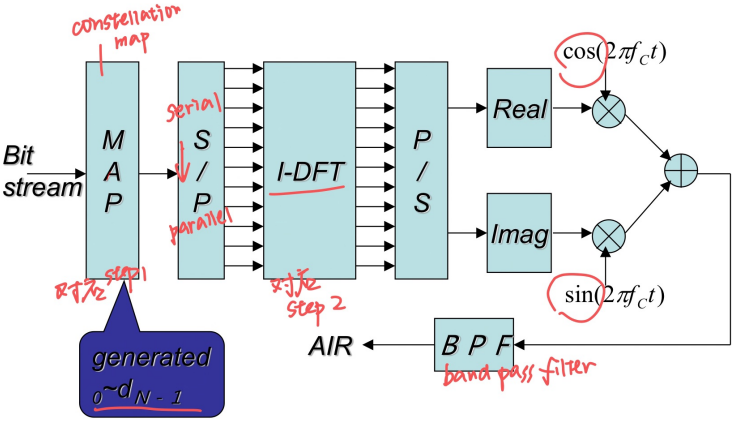
step2: ◆ Perform N times sampling in period T

$$u\left(\frac{k}{Nf_0}\right) = \sum_{n=0}^{N-1} d_n \cdot e^{j2\pi n f_0 \frac{k}{Nf_0}} = \sum_{n=0}^{N-1} d_n \cdot e^{j\frac{2\pi nk}{N}}$$

$$= \sum_{n=0}^{N-1} d_n \cdot \left(e^{j\frac{2\pi}{N}}\right)^{nk} \qquad (k = 0,1,2,\cdots,N-1)$$

$$u(k) = \mathrm{IFFT}(d_n) = \mathrm{IFFT}(a_n + jb_n)$$

OFDM modulator



OFDM解调：

OFDM demodulation

$$s(t)=\sum_{n=0}^{N-1}[a_n\cos\{2\pi(f_c+nf_0)t\}-b_n\sin\{2\pi(f_c+nf_0)t\}]$$ 个组合

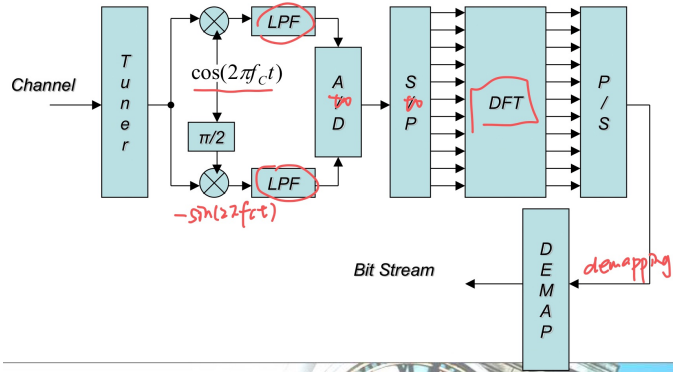
$$LPF[s(t)\cdot\cos(2\pi f_C t)]=\frac{1}{2}\sum_{n=0}^{N-1}\{a_n\cos(2\pi nf_0 t)-b_n\sin(2\pi nf_0 t)\}=\frac{1}{2}s_I(t)$$

$$LPF[s(t)\cdot\{-\sin(2\pi f_C t)\}]=\frac{1}{2}\sum_{n=0}^{N-1}\{a_n\sin(2\pi nf_0 t)+b_n\cos(2\pi nf_0 t)\}=\frac{1}{2}s_Q(t)$$

$$u(t)=s_I(t)+js_Q(t)=\sum_{n=0}^{N-1}d_n\cdot e^{j2\pi nf_0 t}$$

d_n = FFT(u(k))

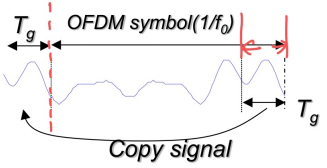
OFDM demodulator



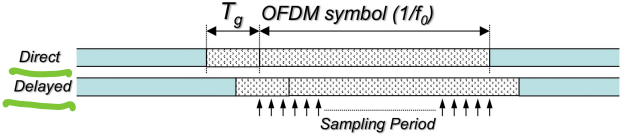
deal with ISI in Multi-path condition

Cyclic Prefix (Guard Interval) T_g

周期前缀



- By adding the Gurard Interval Period, ISI can be avoided



OFDM 特征：

- Feature of OFDM
  1. High Frequency utilization by the square spectrum shape
  2. Multi-path problem is solved by Cyclic Prefix
  3. Multiple services in one OFDM by sharing sub-carriers
  4. Implementation was complicated but NOW possible because of LSI technology progress

OFDM优点：

Advantages of OFDM technologies
- Higher spectral efficiency in real-life time dispersive channels
- More robust – less multi-path interference use guard interval
- Easy to integrate MIMO technologies
- Simpler receiver to cope with real-life time dispersive channels → lower cost

Radio Propagations & Network Architecture

① antenna gain:

$$G_T = \frac{4\pi A}{\lambda^2} = \frac{4\pi f^2 A_e}{c^2}$$

A_e: effective area

f: carrier f

c: 3×10^8

λ: carrier wavelength

② Free space loss

trans… $\frac{P_t}{P_r} = \frac{(4\pi d)^2}{\lambda^2} = \frac{(4\pi f d)^2}{c^2}$, $\lambda = \frac{c}{f}$ (meters)

receiv…

d: propagation distance between antennas (meters)

$L_{dB} = 20\lg\left(\frac{4\pi d}{\lambda}\right)$ dB ($\lg$ = $\log_{10}$)

Noise

① Thermal noise

$N_0 = KT$ W/Hz $\quad K = 1.3803 \times 10^{-23}$ J/K $= -228.6$ dBJ/K

$N = KTB$ Hz bandwidth

$= -228.6$ dBW $+ 10\lg T + 10\lg B$ (dB)

$\frac{E_b}{N_0} = \frac{S}{N_0} = \frac{S}{N_0 R} = \frac{S}{KTR}$ (S: received signal level) 每比特的信号能量与每Hz噪声功率密度之比

$= S_{dBW} - 10\log R - 10\log k - 10\log T$ (dBW)

($\log$ = $\log_{10}$)

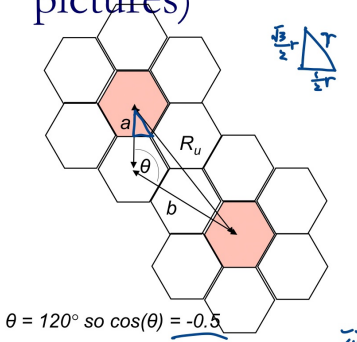
cluster size:

bigger: lower interference. lower capacity. $B = \frac{F_A}{N}$ → total spectrum

B: bandwidth available in cell

当N=7时的蜂窝

Derivation example (N=7 in pictures)



θ = 120° so cos(θ) = -0.5

In general

a is distance between *i* cells

$a = ir\sqrt{3}$

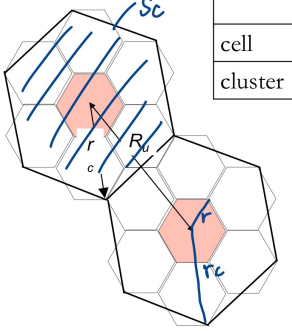
b is distance between *j* cells

$b = jr\sqrt{3}$

$R_u^2 = i^2r^2 3 + j^2r^2 3 + 2\times 0.5ijr^2 3$

法1: $R_u = \left(\sqrt{i^2 + j^2 + ij}\right)\left(r\sqrt{3}\right)$

Cluster radius



	radius	area
cell	r	s
cluster	r_c	s_c

$s_c = Ns$ (N个六边形 area)

$\frac{r_c^2 3\sqrt{3}}{2} = N\frac{r^2 3\sqrt{3}}{2}$

$r_c = r\sqrt{N}$

$R_u = r_c\sqrt{3}$

法2: $R_u = r\sqrt{3N}$

Thick lines define a cluster hexagon of same area as *N* cells

Possible values of N

$R_u = \left(\sqrt{i^2 + j^2 + ij}\right)\left(r\sqrt{3}\right)$ and $R_u = r\sqrt{3N}$

$\left(\sqrt{i^2 + j^2 + ij}\right)\left(r\sqrt{3}\right) = r\sqrt{3N}$ or $\sqrt{i^2 + j^2 + ij} = \sqrt{N}$

$i^2 + j^2 + ij = N$

These means that only certain values of N are possible

i	j	N
1	1	3
1	2	7
2	2	12
1	3	13
2	3	19
1	4	21
3	3	27
2	4	28

∴ $R_u = r\sqrt{3N}$ received signal power: $P \propto 1/r^4$

$\left(P_i = \frac{bP}{d^4} = \frac{bP}{R_u^4} = \frac{bP}{r^4 9N^2}\right)$

$P = \frac{P_0}{(r\sqrt{3N})^4}$

$\frac{P_7}{P_3} = \left(\frac{3}{7}\right)^2 = \frac{9}{49} = -7.4\,dB$